Problem Descriptions

1. **Randomized robotic sensor direction.** A robotic inspection system is used to monitor a circular storage facility. During each scan, its sensor points in a direction A measured in radians clockwise from a fixed reference marker, with $A \in [0, 2\pi)$.

   Because the control system randomly selects the initial direction with no preferred orientation, a circular-uniform model is appropriate. The direction 2π is equivalent to 0 because they represent the same physical orientation.

   The engineering team is particularly interested in directions close to the reference marker. Determine the complete periodic probability density rule for the sensor direction, including what happens outside the stated cycle. Then find the probability that the sensor direction lies within $\frac{\pi}{3}$ radians of the reference marker, interpreting the interval around 0 periodically.

2. **Periodic triangular server-load peak time.** A university data center monitors the time T at which its computing demand reaches its daily peak. The time is measured in hours on $[0, 24)$, with midnight represented by both 0 h and 24 h. Therefore, time of day is periodic with period 24 h.

   For this planning model, the server load is least likely at midnight. The density increases linearly from zero at midnight to its maximum at 08:00, then decreases linearly back to zero at the next midnight. A circular-triangular pdf represents this daily pattern.

   The data-center team is interested in peak times close to midnight. Determine the complete periodic probability density rule for the peak time, including what happens outside the displayed day. Then find the probability that the daily peak occurs within 4 hours of midnight, interpreting the interval around midnight periodically.

Equations Needed

PDF:

$$f(x) \geq 0, \qquad \int f(x)\,dx = 1,$$

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

Uniform:

$$h = \frac{1}{L}.$$

Line:

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \qquad y = mx + b.$$

Triangle:

$$\frac{1}{2}BH = 1, \qquad H = \frac{2}{B}.$$

$$m_1 = \frac{H}{b-a}, \qquad m_2 = -\frac{H}{c-b}.$$

Periodic:

$$f(x + P) = f(x).$$

$$[-a, a] \to [0, a] \cup [P - a, P).$$

Solutions

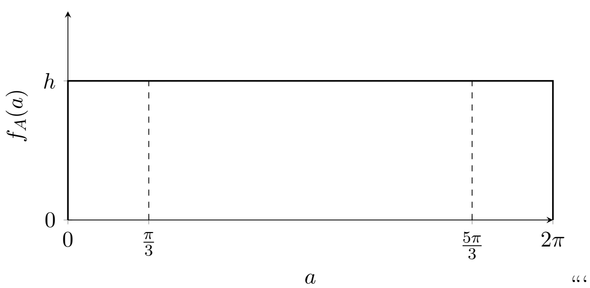
1 Periodic uniform distribution

“‘

1.1 Support and Shape

The random variable A represents the robotic sensor direction in radians. One complete cycle is represented by $[0, 2\pi)$, and the direction is periodic because a full rotation returns the sensor to the same physical orientation. The controller selects all directions with equal likelihood, so the density is a rectangle of unknown height h over the complete cycle and is zero outside the displayed support.

Shape from the context



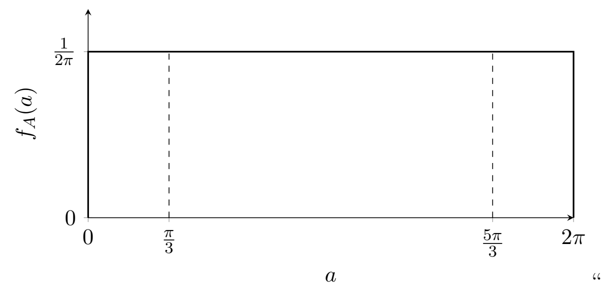
“‘

1.2 Total Area = 1

The rectangular pdf must have total area 1 over one complete cycle:

$$1 = (2\pi)h, \qquad h = \frac{1}{2\pi}.$$

Completed periodic density graph



“‘

“‘

1.3 Straight Line Rule

The density is constant throughout the support, so the graph is horizontal. Therefore its slope is $m = 0$, and the constant rule is $f_A(a) = \frac{1}{2\pi}$ on the support. The periodic nature of the direction is expressed by $f_A(a + 2\pi) = f_A(a)$.

1.4 Complete pdf

The complete pdf can be written on one displayed cycle as $f_A(a) = \begin{cases} \dfrac{1}{2\pi}, & 0 \le a < 2\pi, \\ 0, & \text{otherwise.} \end{cases}$

“‘ The displayed endpoints 0 and 2π represent the same physical direction, and the periodic extension satisfies

$$f_A(a + 2\pi) = f_A(a).$$

“‘

1.5 Interval Around the Reference Marker

The interval is

‘‘‘

$$-\frac{\pi}{3} \leq A \leq \frac{\pi}{3} \qquad \longrightarrow \qquad \left[0, \frac{\pi}{3}\right] \cup \left[\frac{5\pi}{3}, 2\pi\right).$$

The two pieces have total width

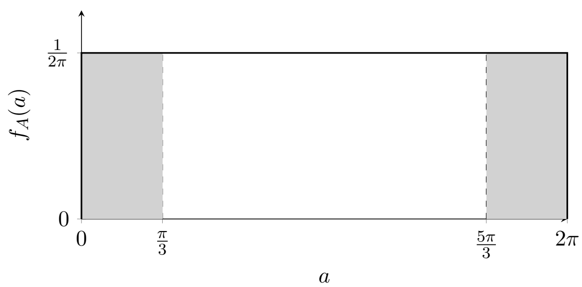
$$\frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}.$$

Therefore,

$$P\left(-\frac{\pi}{3} \leq A \leq \frac{\pi}{3}\right) = \frac{2\pi}{3}\frac{1}{2\pi} = \frac{1}{3} = 33.33\%.$$

Thus, approximately 33.3% of randomized sensor directions lie within $\frac{\pi}{3}$ radians of the reference marker.

Periodic interval around 0



‘‘‘

‘‘‘

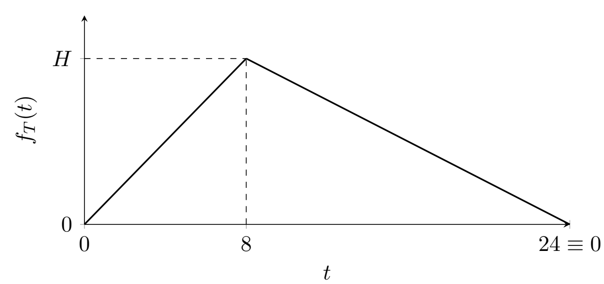
2 Periodic triangular distribution

‘‘‘

2.1 Support and Shape

The random variable T represents the time of the daily server-load peak in hours. One complete cycle is represented by $[0, 24)$, and the period is 24 h because the clock repeats every day. The density is zero at midnight, rises steadily to its maximum at $t = 8$, and then falls steadily to zero at the next midnight. Therefore, the graph is a triangle with unknown maximum height H and mode at $t = 8$.

Shape from the context



‘‘‘

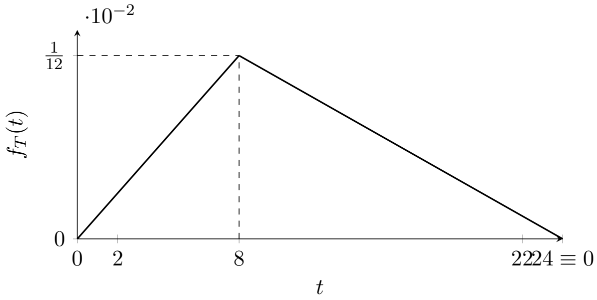
2.2 Total Area = 1

The triangular pdf must have total area 1:

$$1 = \frac{1}{2}(24)H, \qquad H = \frac{1}{12}.$$

The completed triangle therefore has maximum height 1/12 at $t = 8$.

Completed periodic density graph



“‘

“‘

2.3 Straight Line Rules

The rising section connects $(0,0)$ and $\left(8, \frac{1}{12}\right)$:

“‘

$$(0,0), \left(8, \frac{1}{12}\right) \qquad \longrightarrow \qquad m = \frac{\frac{1}{12} - 0}{8 - 0} = \frac{1}{96}, \quad b = 0 \qquad \longrightarrow \qquad f_T(t) = \frac{t}{96}.$$

The falling section connects $\left(8, \frac{1}{12}\right)$ and $(24, 0)$:

$$\left(8, \frac{1}{12}\right), (24,0) \qquad \longrightarrow \qquad m = \frac{0 - \frac{1}{12}}{24 - 8} = -\frac{1}{192} \qquad \longrightarrow \qquad f_T(t) = \frac{24 - t}{192}.$$

“‘

“‘

2.4 Complete pdf

The complete pdf on one displayed day is

$$f_T(t) = \begin{cases} \dfrac{t}{96}, & 0 \le t \le 8, \\ \dfrac{24 - t}{192}, & 8 < t < 24, \\ 0, & \text{otherwise.} \end{cases}$$

The periodic extension is

$$f_T(t + 24) = f_T(t).$$

The two triangular sections have areas

$$\frac{1}{2}(8)\frac{1}{12} + \frac{1}{2}(16)\frac{1}{12} = \frac{1}{3} + \frac{2}{3} = 1,$$

confirming that the pdf is correctly normalized. “‘

2.5 Interval Around Midnight

The interval is

“‘

$$-4 \le T \le 4 \qquad \longrightarrow \qquad [0,4] \cup [20,24).$$

For $[0,4]$:

$$f_T(4) = \frac{4}{96} = \frac{1}{24} \qquad \longrightarrow \qquad A_1 = \frac{1}{2}(4)\frac{1}{24} = \frac{1}{12}.$$

For $[20,24)$:

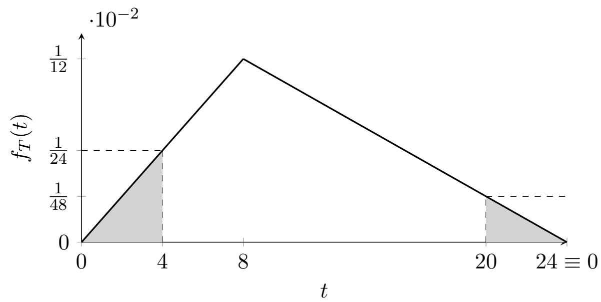
$$f_T(20) = \frac{24-20}{192} = \frac{1}{48} \qquad \longrightarrow \qquad A_2 = \frac{1}{2}(4)\frac{1}{48} = \frac{1}{24}.$$

Therefore,

$$P(-4 \leq T \leq 4) = \frac{1}{12} + \frac{1}{24} = \frac{1}{8} = 12.5\%.$$

Thus, 12.5% of days have their modeled server-load peak within 4 hours of midnight.

Periodic interval around midnight



‘‘‘